INTRODUCTION

Earlier lessons demonstrated the relationship between bond prices and yields-to-maturity (YTMs), as well as other features, such as coupon rate and time-to-maturity. Two important considerations for interpreting and determining yields-to-maturity and other yield measures are the assumed frequency of compounding interest and the presence of embedded options that could affect cash flow amounts or timing. This module explores these considerations and extends the analysis of yields by introducing spread measures, which compare yields to benchmark rates to ascertain how much an investor would be compensated for taking certain risks.

Most of the examples and exhibits used throughout the reading can be downloaded as a Microsoft Excel workbook. Each worksheet in the workbook is labeled with the corresponding example or exhibit number in the text.

Category
Yield and Yield Spread Measures for Fixed-Rate Bonds

Learning Module Overview

- Yields-to-maturity allow analysts to use a single measure to compare bonds with varying maturities and coupons. An important factor in yield determination and interpretation is the frequency of compounding, known as periodicity.
- All else equal, more frequent compounding results in greater future values or returns. Therefore, compounding more frequently at a lower rate can be equivalent to compounding less frequently at a higher rate; the more frequent compounding makes up for the lower rate.
- Prices and yields of bonds with embedded options require adjustments to reflect the value of the option. The option-adjusted yield spread and yield-to-worst are measures that reflect option values.
- Yield measures for a bond, as well as changes in them, can be decomposed into a benchmark rate and a spread over the benchmark rate. Benchmark rates reflect macroeconomic, “top-down” conditions that affect all bonds in a market, while spreads capture issuer-specific, “bottom-up” factors, such as credit risk, liquidity, and taxation.
- Similar to how a yield curve graphically depicts the relationship between yields-to-maturity and times-to-maturity for securities with the same risk profile, benchmark rates and spreads can be graphed by time-to-maturity as well, to show the term structure of credit spreads.

Learning Module Self-Assessment

These initial questions are intended to help you gauge your current level of understanding of the learning module.

1. Complete the sentences by filling in the blanks using the following words:

   annualized

   compounded

   standardized

not _____.

Hide Solution

Solution:

Both capital market securities' yields and money market rates with maturity dates shorter than one year are annualized. However, money market rates are not compounded.

2. A three-year sovereign non-callable bond is priced at 106.24 per 100. The bond pays a 2% semiannual coupon. The annual yield-to-maturity for the bond is closest to:

A. 0.077%.
B. 0.039%.
C. 1.883%.

Hide Solution

Solution:

A is correct. Given that the bond pays coupons semiannually, the semiannual yield-to-maturity is calculated by solving for *r* in the following equation. Additionally, the IRR or YIELD functions in Microsoft Excel and Google Sheets can be used.

$$106.24 = \frac{1}{(1+r)^1} + \frac{1}{(1+r)^2} + \frac{1}{(1+r)^3} + \frac{1}{(1+r)^4} + \frac{1}{(1+r)^5} + \frac{101}{(1+r)^6}.$$
$$r = -0.00038596.$$

Given that the yield is expressed in semiannual terms, it needs to be annualized by multiplying by 2, which results in –0.000772, or –0.077%.

B is incorrect because it is the semiannual, not annual, yield.

C is incorrect because 1.883% is the bond's current yield, not yield-to-maturity.

3. A fixed-income analyst obtains the following data for three corporate bonds with the same maturity dates. The relevant government bond benchmark rate, calculated on a semiannual bond equivalent basis, is 1.10%. Calculate the G-spread for each bond.

	Bond A	Bond B	Bond C
Yield-to-maturity	1.271%	1.213%	1.178%
Coupon frequency and assumed periodicity	Annual	Semiannual	Quarterly

Hide Solution

Solution:

The G-spread for Bond B is (0.01213 – 0.011) = 11.3 bps.

Before computing the G-spreads for Bonds A and C, the yields-to-maturity must first be converted to a periodicity of two. The semiannual equivalent yield for Bond A is 1.267%:

$$\left(1 + \frac{APR_m}{m}\right)^m = \left(1 + \frac{APR_n}{n}\right)^n.$$
$$\left(1 + \frac{0.01271}{1}\right)^1 = \left(1 + \frac{APR_2}{2}\right)^2.$$
$$APR_2 = 0.01267 = 1.267\%.$$

Thus, its G-spread is (0.01267 – 0.011) = 16.7 bps.

The semiannual equivalent yield for Bond C is 1.1797%:

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$$\left(1+\frac{\quad}{4}\right) = \left(1+\frac{\quad}{2}\right).$$

$$APR_2 = 0.011797 = 1.1797\%.$$

Thus, its G-spread is (0.011797 – 0.011) = 8 bps.

4. If a bond has an embedded call option and the value of the call option is positive, the option-adjusted price, compared to the flat price of the bond, is most likely:

A. lower.
B. the same.
C. higher.

Hide Solution

Solution:

A is correct. The value of the call option is the price of the option-free bond minus the price of the callable bond. If the value of the call option is positive, then the option-adjusted price is lower compared to the flat price of the bond.

B is incorrect because if the value of the call option is positive, the option-adjusted price compared to the flat price of the bond is lower, not the same.

C is incorrect because if the value of the call option is positive, the option-adjusted price compared to the flat price of the bond is lower, not higher.

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Discussion

What do you want to discuss... (required)

DS

Question 3 decimal error for bond A

Thus, its G-spread is (0.01267 – 0.011) = 16.7 (not 167)

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in Question 4.Isn't the option-adjustic price equal to the flat price plus the price of call option ? In that case, the option-adjustic price should be higher than the flat price. Why not choose C?

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BA

In Question 2. I get that the answer is -0,077%, yet I don´t understand why the alternative is A. 0,077% without the minus sign. Is this a typo? Or it does not actually matter in this context whether the YTM is positive or negative?

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